

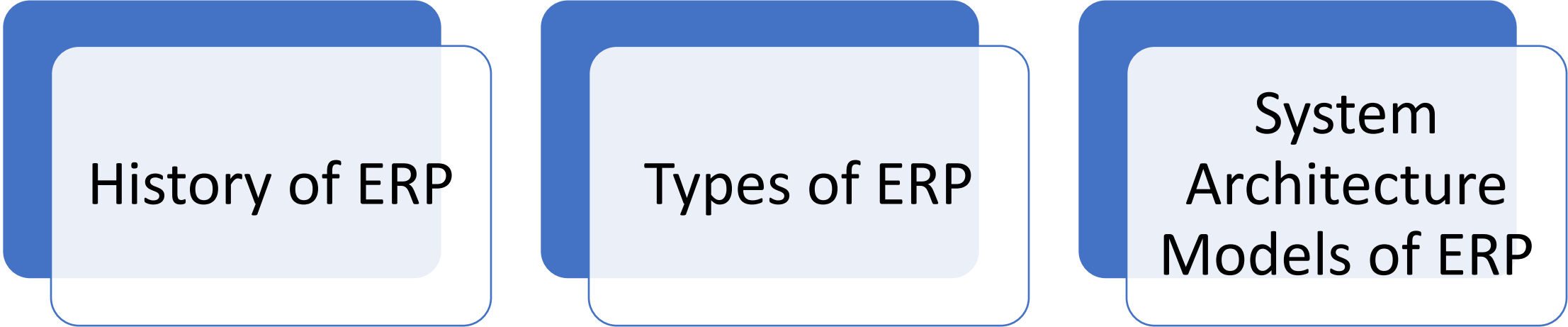
ERP

The Development of ERP Systems



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OUTLINE



History of ERP

Types of ERP

System
Architecture
Models of ERP

Learning Objectives

01

- Understand the history and evolution of ERP

02

- Describe the types of ERP

03

- Explain the types of system architecture for ERP

Recap

Enterprise Resource Planning (ERP)

- Enterprise Resource Planning (ERP) programs are **software used by companies to manage information in every area of the business.**
- ERP programs help **manage company-wide business processes using a common database and shared management reporting tools.**
- ERP software supports the efficient operation of business processes by **integrating activities throughout a business.**

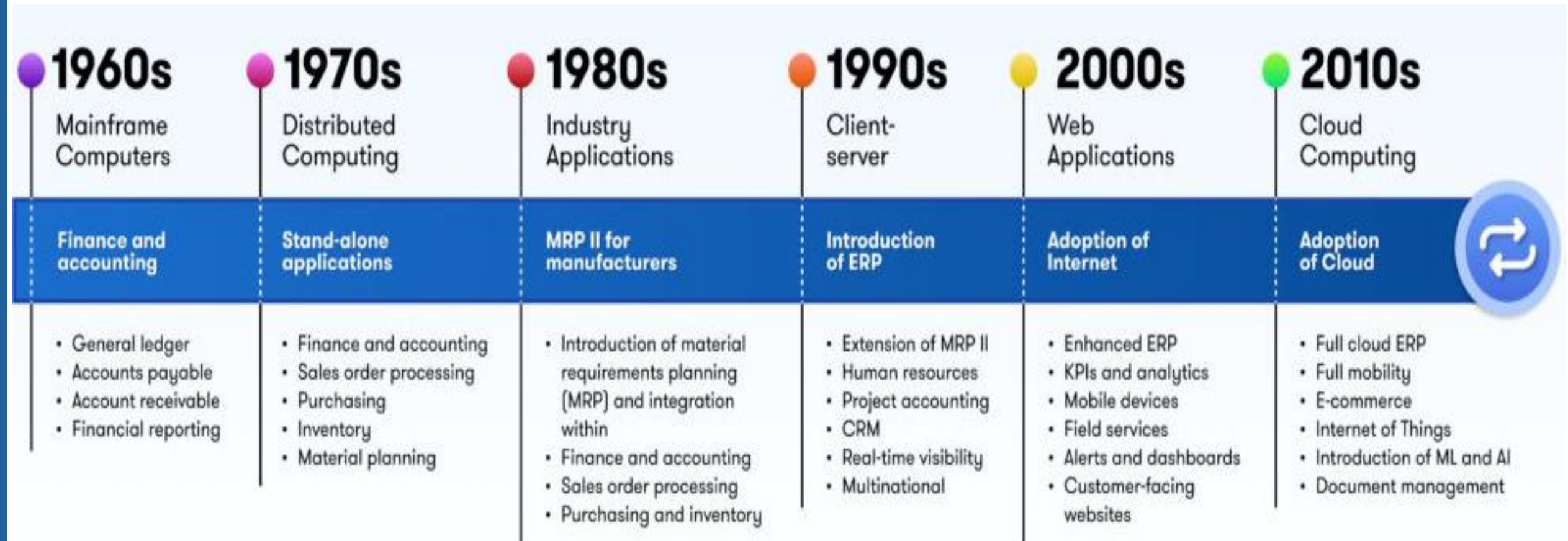


Business Function vs Business Process

Aspect	Business Function	Business Process
Focus	Department or role	Workflow or activity flow
Boundaries	Usually stays within one department	Often spans multiple departments
Goal	Performs specialized tasks	Achieves a business outcome
Example	HR department managing employee records	Onboarding a new employee (HR + IT + Payroll)

History of ERP

The History of ERP



The History of ERP - 1960s: Early beginnings



The term ERP was first used in the 1990s by the Gartner Group, but enterprise resource planning systems actually have their roots deep in the manufacturing industry and can trace their history back to the 1960s.

At this time, manufacturers needed a better way to manage, track, and control their inventory. Basic software solutions, known as MRPs or Material Requirements Planning systems, were developed to meet their needs. These systems helped manufacturers monitor inventory and reconcile balances, as well as included very basic manufacturing, purchasing, and delivery functions.

The History of ERP - 1970s and 1980s: MRP systems



Through the 1970s, more and more manufacturers started to adopt MRP systems, and the systems themselves became more sophisticated.

The History of ERP



1980s
Industry Applications

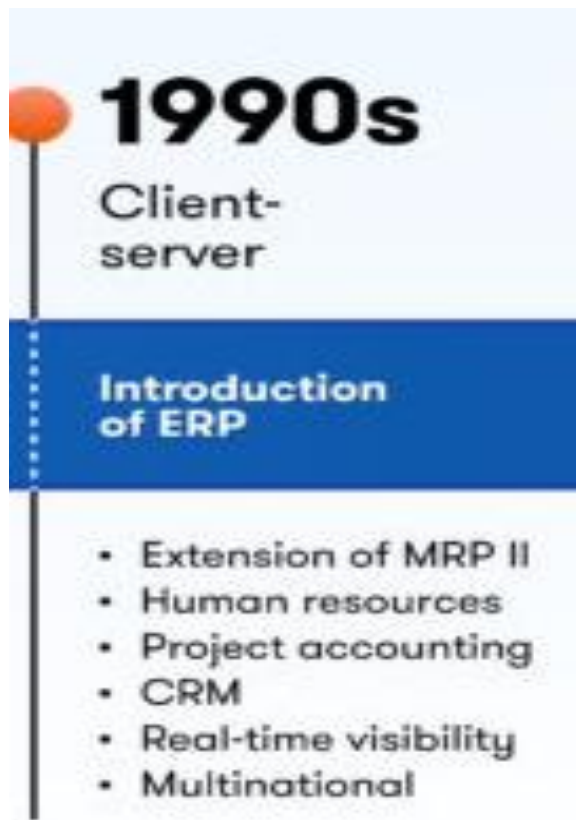
MRP II for manufacturers

- Introduction of material requirements planning (MRP) and integration within
- Finance and accounting
- Sales order processing
- Purchasing and inventory

By the 1980s, MRP systems evolved into what became known as MRP II or Manufacturing Resource Planning systems.

More manufacturing processes were added to the original MRP systems, and these MRP II systems had expanded capabilities and were better able to handle scheduling and production processes.

The History of ERP - 1990s and beyond: True ERPs



In the 1990s, the first true ERP systems came into use. These systems further expanded beyond previous iterations' basic inventory control and manufacturing processes to include other departments and functions, such as accounting, finance, and sales. These systems set the stage for ERP solutions as we've come to know them today, by integrating multiple processes and departments into one system.

The History of ERP - 2000s: ERP 11



With the rapid technological developments of the early aughts, ERP software solutions could send and receive information from external organizations, vendors, and clients. The term ERP II arose in 2000 to describe these new advancements.

ERP implementation now resulted in three main functionalities:

- Facilitate operations between different organizations
- Communicate with vendors and partners throughout the supply chain
- Provide transparency to clients about inventory and shipping info

Customer relationship management (CRM) and business intelligence (BI) add-ons allowed manufacturers to nurture customer relationships and evaluate internal and external data to inform business functions at every level.

The History of ERP - 2010s: Big Data and Artificial Intelligence



Over the next few decades, ERP systems became more advanced. In recent years, big data and AI have once again transformed ERP systems, making them more powerful and adaptive. Big data enables ERP systems to handle massive amounts of information, providing businesses with valuable insights into their operations. AI adds a layer of intelligence, allowing ERP systems to learn from data patterns and make informed predictions, and AI-driven automation streamlines repetitive tasks.

The integration of big data and AI in ERP systems has ushered in a new era of agility and intelligence, empowering manufacturers to navigate the complexities of the modern business landscape.

Types of ERP

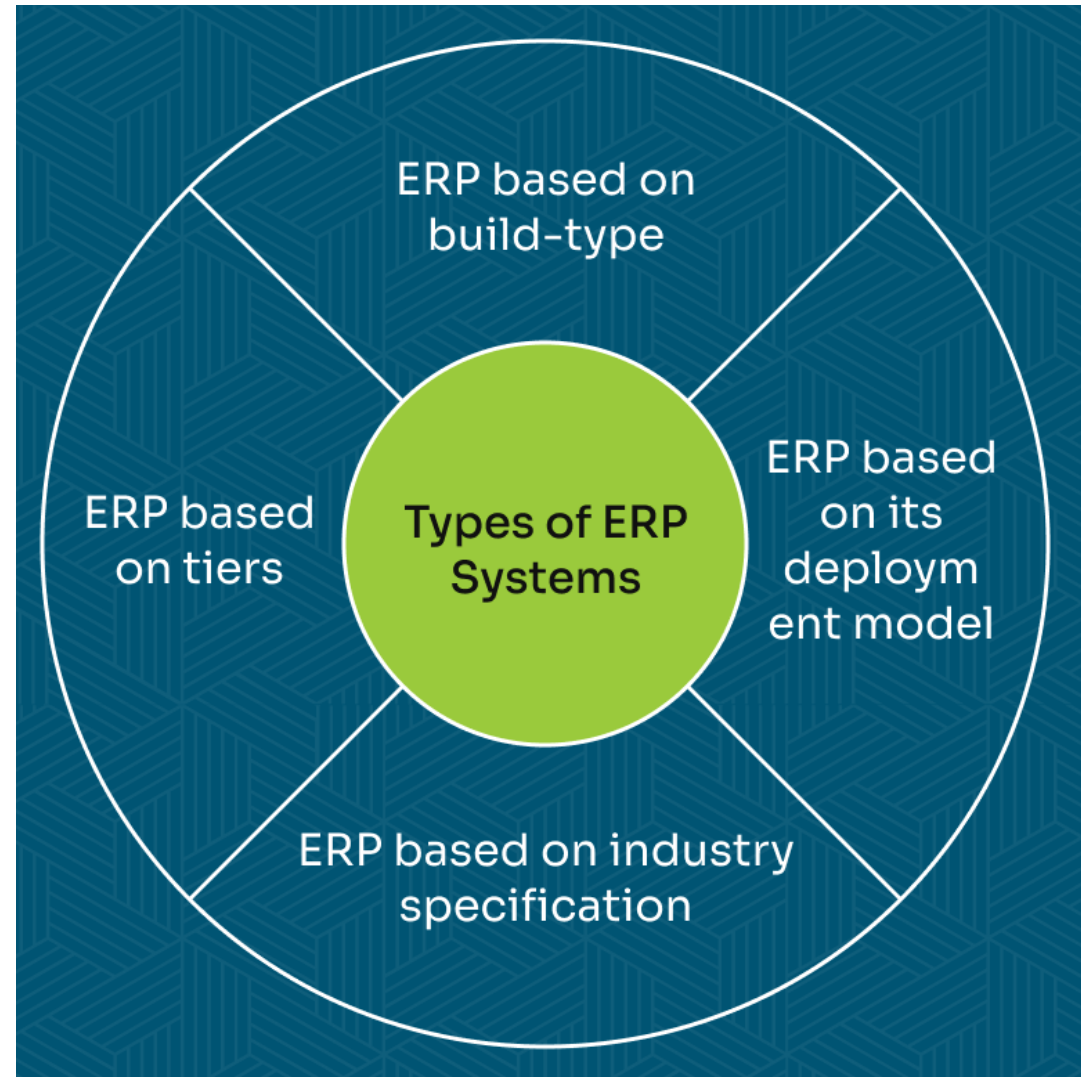
Types of ERP

- How to select the right ERP for the business
 - Choosing new ERP or Upgrade existing systems
- Business must have a clear vision of the system requirements and they will align with internal business processes
- Every business has unique challenges and needs.
- Choosing the wrong ERP system or partner can lead to implementation issues:
 - Poor return on investment (ROI)
 - Budget overruns
 - Underutilized systems



ERP?

Types of ERP



Types of ERP – Build Type

Custom-Built ERP

- Tailor-made ERP system developed in-house or by third-party developers
- Designed specifically for the unique processes and workflows of an organization
- Offers full control over features, interface, and integration
- Often used by organizations with specialized needs or legacy systems

Ready-made ERP

- Off-the-shelf ERP solutions provided by software vendors
- Includes standard modules for accounting, HR, inventory, etc.
- Quick to deploy with predefined best practices
- Suitable for most industries with general business needs

Types of ERP – Build Type

	Ready-Made Software	Custom Software
Advantages	• Quick implementation • Easy to adopt • Low upfront cost • Maintenance by provider	• Totally custom to YOU • Total ownership • Flexible over time to your requirements • More integration options • Greater long-term ROI
Disadvantages	• Little to no customisation • Features not required • Quickly outgrown • Less integration options	• High upfront investment • More time required for development process

Types of ERP – Deployment Model



On-premises

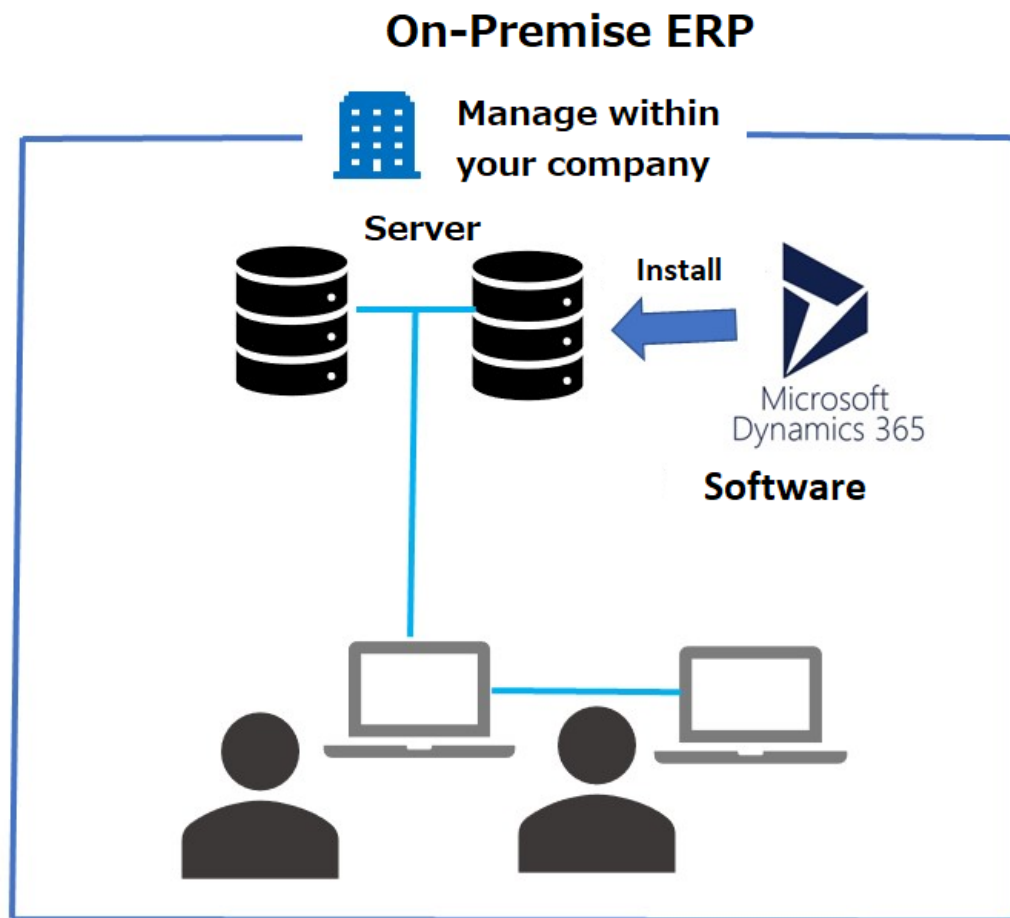


Cloud



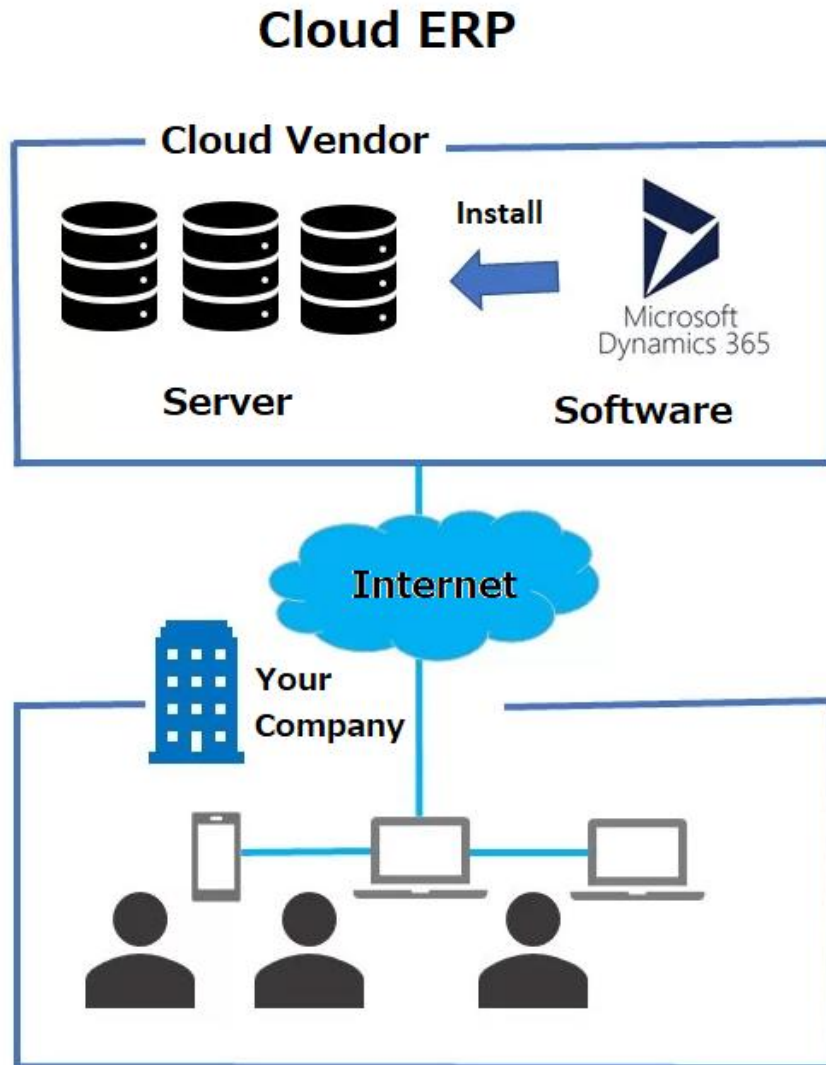
Hybrid

Types of ERP – Deployment Model (On-Premises)



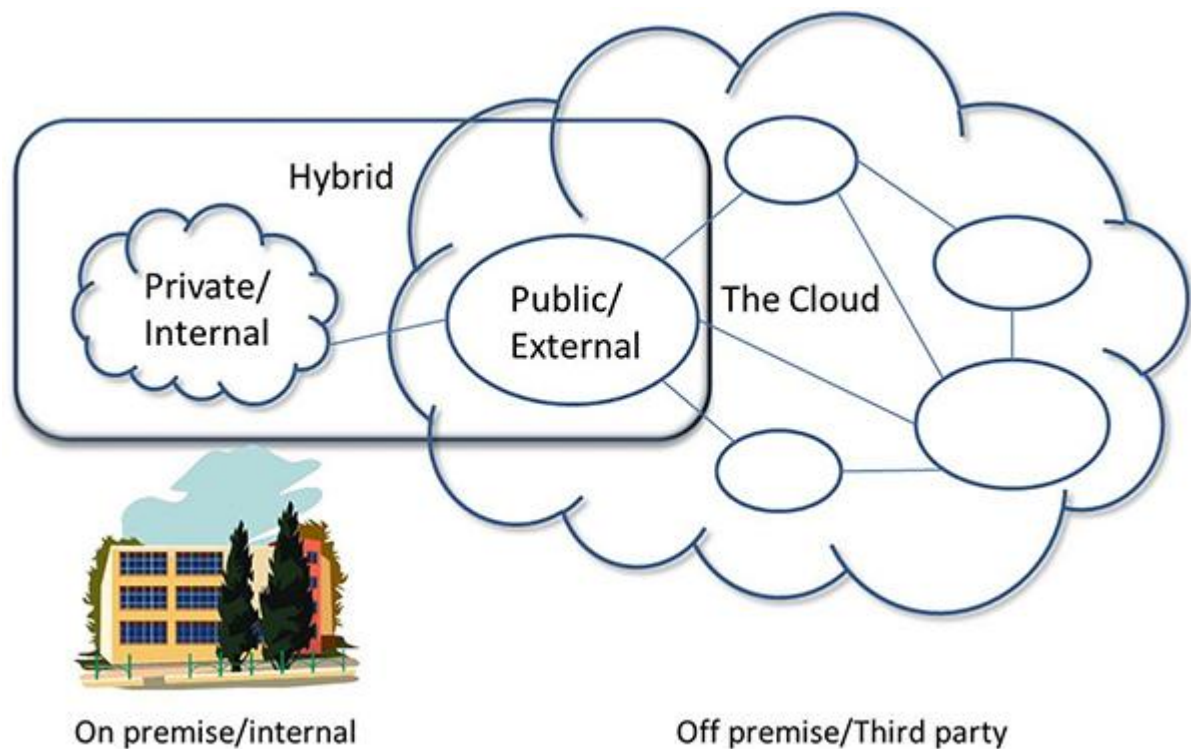
- ERP software installed and run on company-owned hardware and servers
- Full control over system security, updates, and infrastructure
- Typically managed by in-house IT team
- Higher upfront cost for licenses and infrastructure

Types of ERP – Deployment Model (Cloud-based)



- ERP software hosted on the vendor's cloud infrastructure
- Accessed through the internet via web browser or mobile app
- Vendor handles updates, maintenance, and security
- Subscription-based pricing (SaaS model)

Types of ERP – Deployment Model (Hybrid)



- Combines both on-premise and cloud-based components

Core functions may be on-premise, while others (e.g., analytics or mobile access) run in the cloud

Offers flexibility and gradual cloud adoption

- Useful for companies with mixed infrastructure needs

Types of ERP – Deployment Model

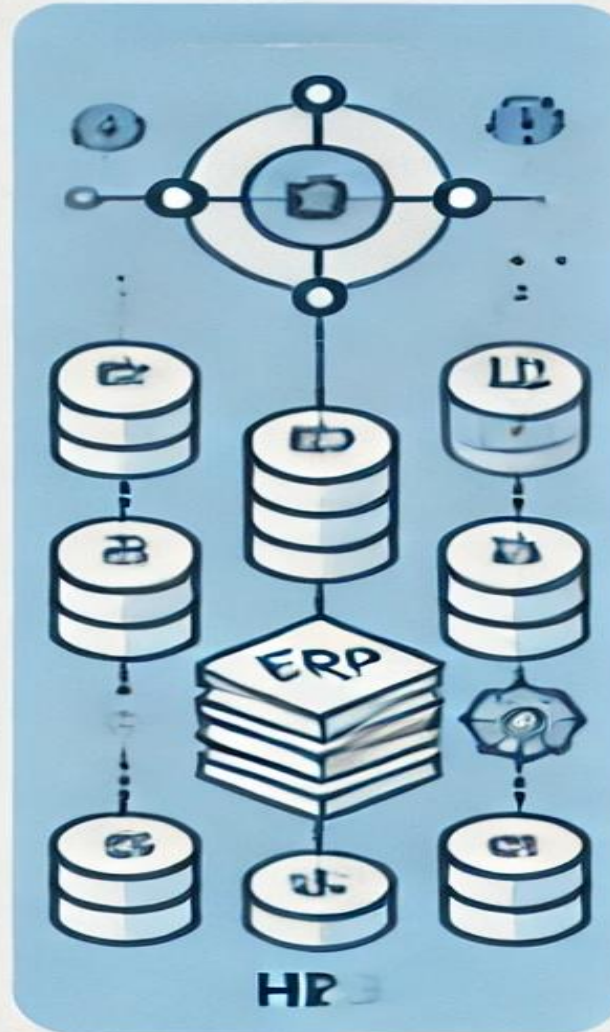
Aspect	On-Premise ERP	Cloud ERP	Hybrid ERP
Deployment	Installed locally on company servers	Hosted on vendor's cloud servers	Mix of on-premise and cloud modules
Cost	High upfront cost for hardware & licenses	Subscription-based, lower initial cost	Mixed cost (licenses + subscription)
Control	Full control over data and systems	Vendor controls infrastructure & updates	Control over on-premise components
Maintenance	Internal IT handles everything	Vendor-managed maintenance & support	Shared responsibility
Scalability	Limited by in-house infrastructure	Highly scalable based on vendor capacity	Scalable through cloud parts
Internet Dependency	Not reliant on internet	Requires stable internet connection	Depends on which parts are cloud-based
Security	Managed internally; may vary by organization	Vendor implements standardized, certified security	Security split across environments
Examples	SAP ECC (on-prem version)	Oracle NetSuite, SAP S/4HANA Cloud	Microsoft Dynamics 365 (with hybrid deployment)

Types of ERP – Industry Specification

Core ERP



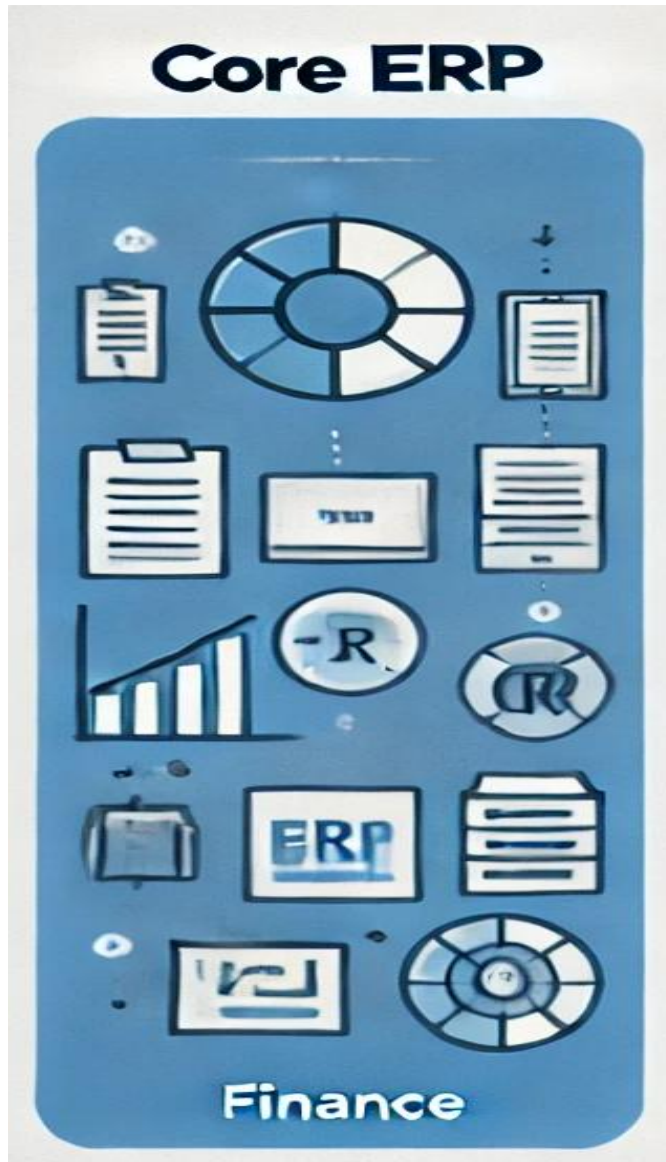
Extended ERP



Vertical ERP



Types of ERP – Industry Specification (Core ERP)



Description:

Core ERP systems offer fundamental functionalities essential for running a business. These systems typically include modules for MASH – Marketing, Accounting & Finance, Supply Chain Management (SCM), Human resources (HR), and customer relationship management (CRM).

Why It Fits Basic Business Needs:

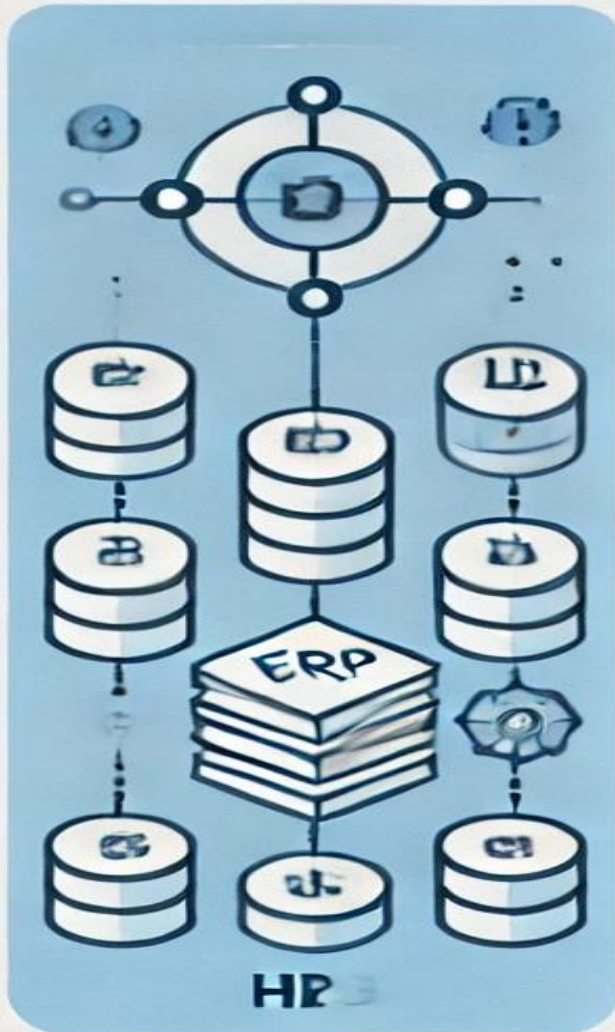
- **Fundamental Operations:** Core ERP systems provide the essential tools for day-to-day operations, ensuring the efficient management of critical business processes.
- **Cost-Effective:** These systems are often more affordable, making them suitable for businesses that need essential features without extensive customization.
- **Ease of Implementation:** With a focus on core functionalities, these systems can be implemented more quickly and with less complexity.

Real-Life Example:

Small Manufacturing Company: A small manufacturing company might use an ERP system like Odoo, which offers core functionalities such as finance, HR, and CRM. This allows the company to manage its financial transactions, employee records, and customer relationships without advanced features.

Types of ERP– Industry Specification (Extended ERP)

Extended ERP



Description:

Extended ERP systems build on core functionalities by adding advanced capabilities such as analytics, business intelligence (BI), supply chain management (SCM), and project management.

Why It Fits Growing Businesses:

- **Advanced Features:** These systems offer additional modules that support more complex and strategic business processes.
- **Scalability:** They can scale with the business as it grows, adding new functionalities to meet evolving needs.
- **Improved Decision-Making:** These systems' enhanced analytics and BI provide deeper insights into business performance, aiding in better decision-making.

Real-Life Example:

Mid-Sized Retail Chain: A mid-sized retail chain might use Netsuite for its extended ERP capabilities, including supply chain management and advanced analytics. This helps the company optimize inventory levels, manage supplier relationships, and gain insights into sales trends, ultimately improving operational efficiency and profitability.

Types of ERP - Industry Specification (Vertical ERP)

Vertical ERP



Description:

Vertical ERP systems are tailored to meet the specific needs of particular industries. They offer industry-specific features and modules that address unique business requirements.

Why It Fits Industry-Specific Needs:

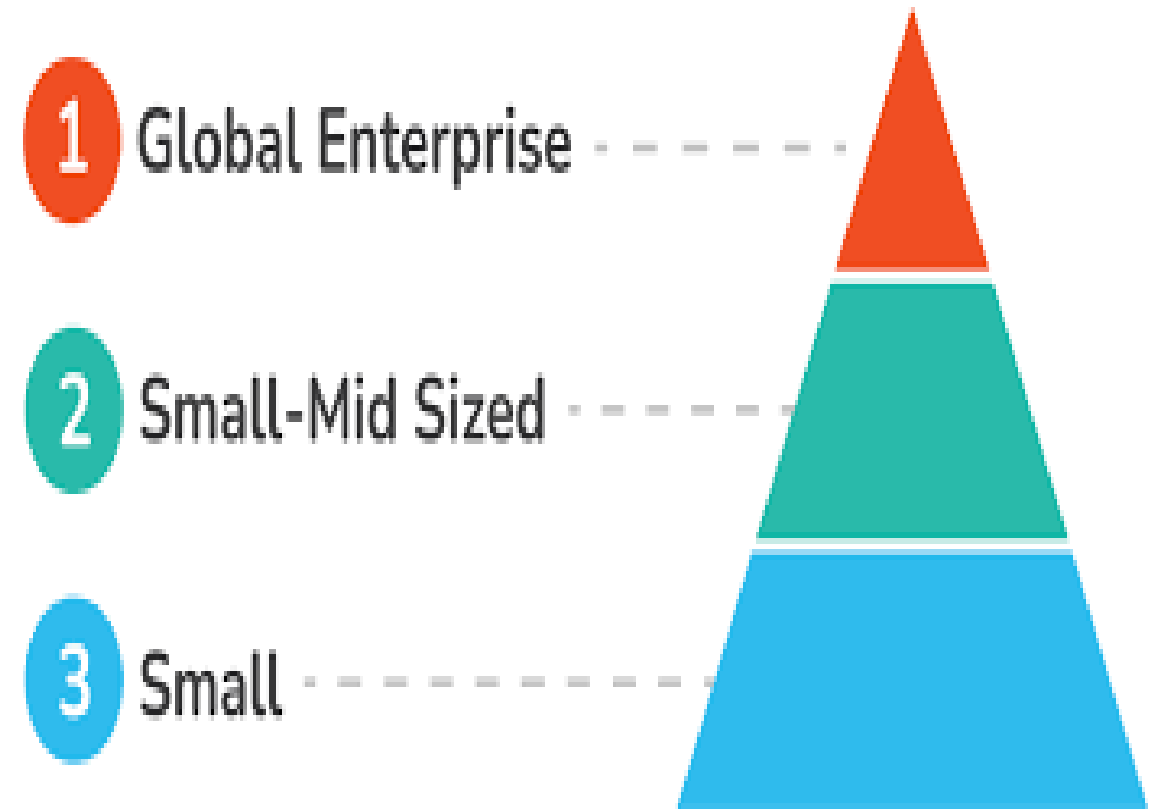
- **Specialized Features:** These systems include functionalities designed to handle certain industries' specific processes and compliance requirements.
- **Optimized Workflows:** Vertical ERP systems streamline industry-specific workflows, improving efficiency and reducing the need for extensive customization.
- **Regulatory Compliance:** They ensure businesses adhere to industry regulations, reducing non-compliance risk.

Real-Life Example:

Healthcare Provider: A large healthcare provider might use an ERP system like Infor Healthcare, which offers industry-specific modules for patient management, electronic medical records, and regulatory compliance. This ensures the provider can manage patient care efficiently while adhering to healthcare regulations.

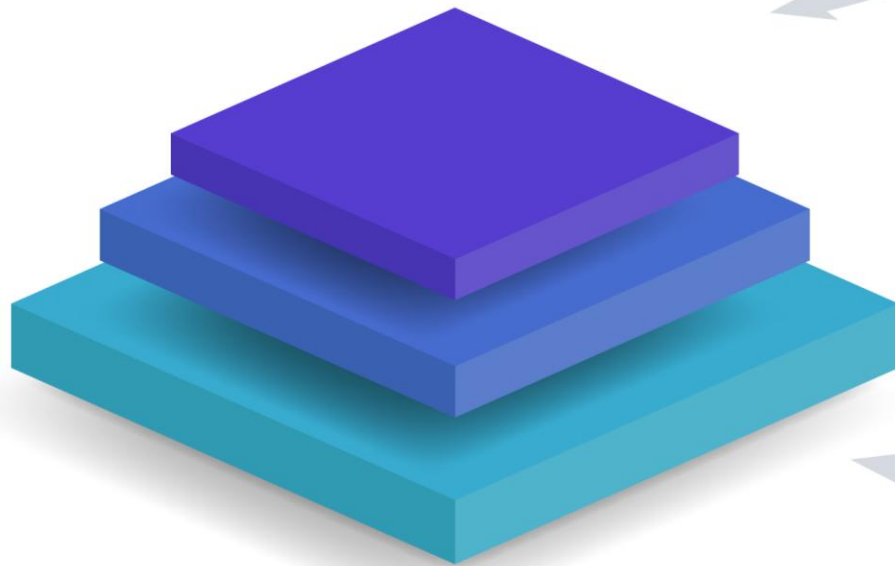
Types of ERP – Tiers

- Separation of ERP systems into classes or tiers is based on a variety of parameters, such as
 - the size of enterprise served (i.e., the number of anticipated users),
 - the type of industry or business category, and
 - the method of deployment for the ERP software.
- To this day, ERP systems are nominally divided into three tiers.



Types of ERP – Tiers

With so many different enterprise resource planning (ERP) software systems available on the market, it's useful to classify them. Let's take a look at tier 1, 2, and 3 ERPs and some popular ERP examples.



Tier 1

Tier 1 solutions are comprehensive systems designed for large, diversified companies with operations on multiple continents.

SAP, Oracle, Microsoft



Tier 2

Solutions in this tier have fewer features and are designed for mid-market businesses, of which there are many more around the world.

Sage, Infor, Epicor



Tier 3

Tier 3 solutions are typically niche or industry-specific solutions that cover small to midsize business (SMB) clients that rarely operate in multiple countries and markets.

xTuple, BizAutomation, Aptean ERP, Expandable ERP

ERP

QUIZ



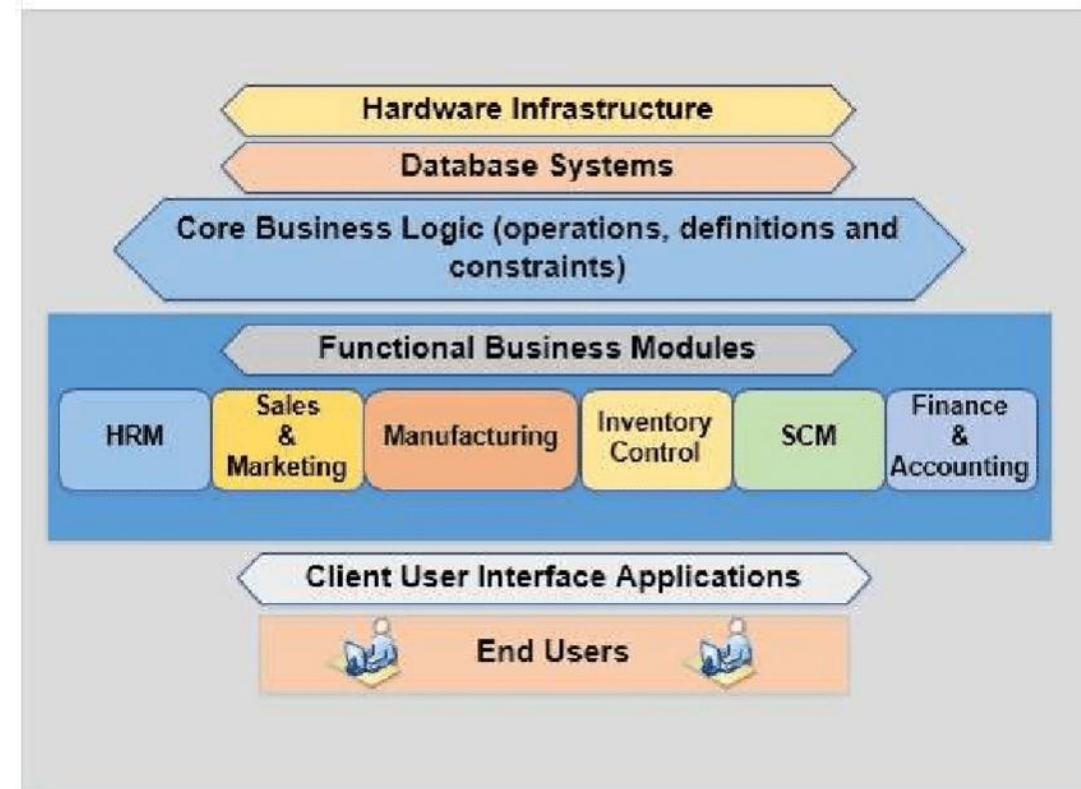
ACTIVITY



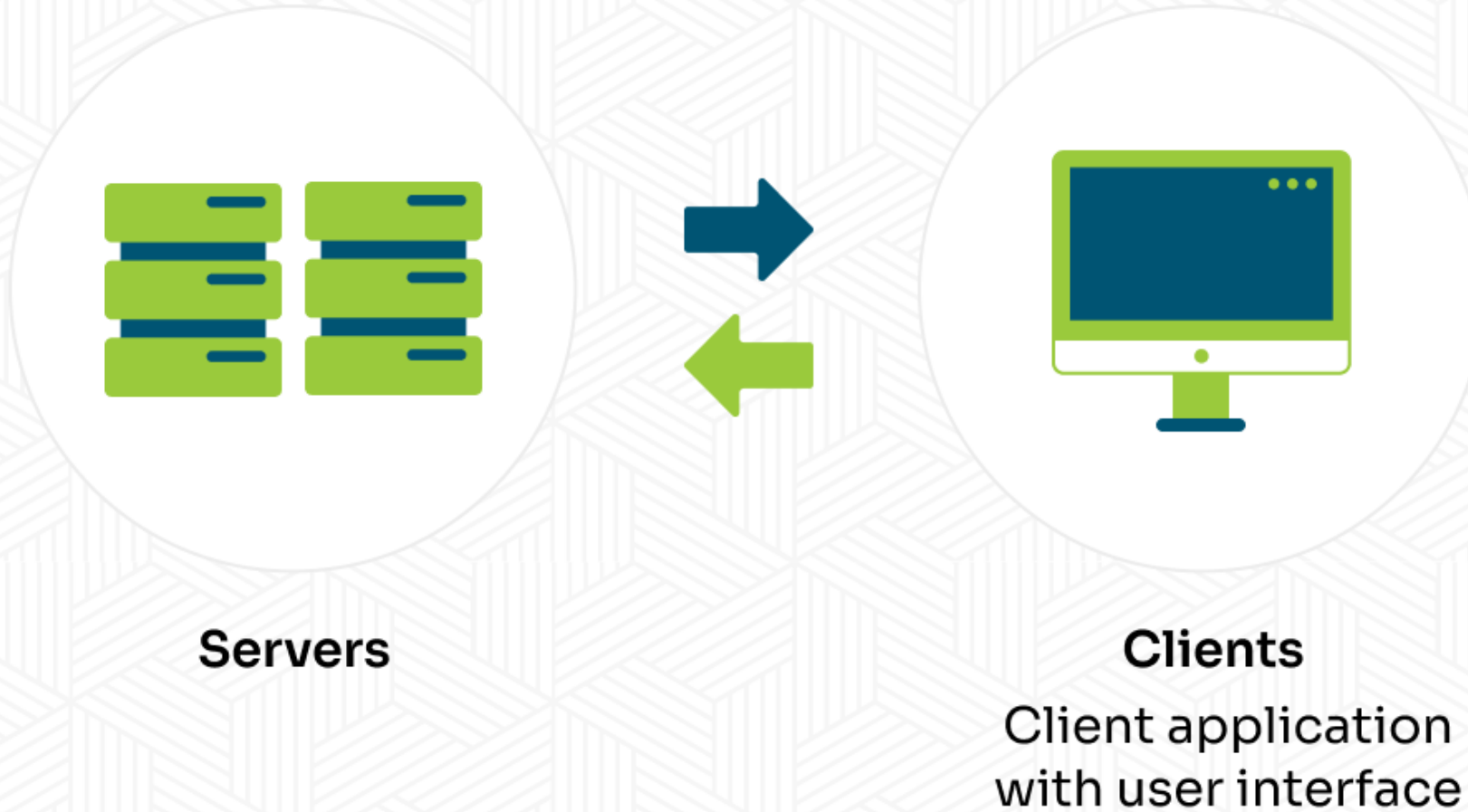
System Architecture Models of ERP

System Architecture Models of ERP

- ERP architecture determines how the modules in an ERP system are connected.
- There are three ERP architecture models:
 - two-tier,
 - three-tier, and
 - Service-Oriented Architecture (SOA).



Two-Tier ERP Architecture



System Architecture Models of ERP – Two-Tier

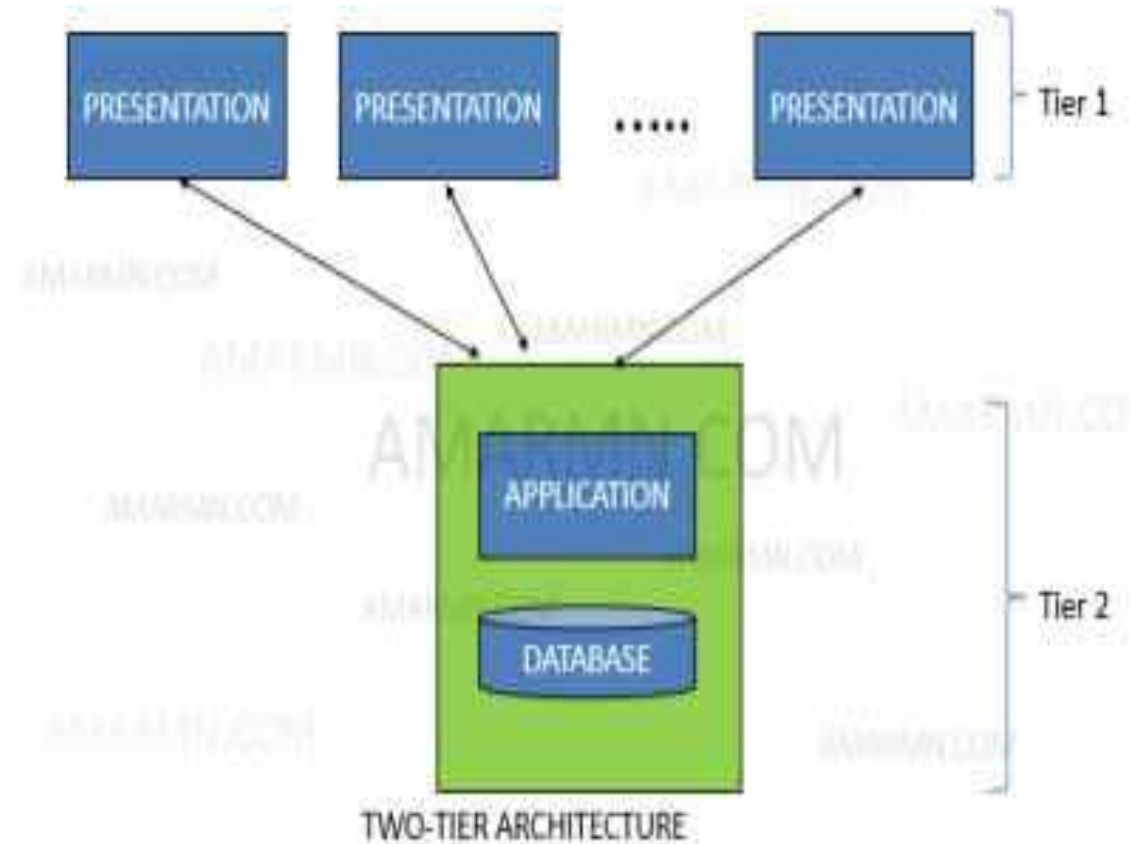
A two-tier ERP architecture is grouped into layers: servers and clients. Servers are typically responsible for handling database requests. The clients include the applications that request the data from the servers and use it for different functional modules.

This model can be implemented with thin clients where the servers take care of both database and application tasks, and the clients are only responsible for providing a user (graphical) interface to the end user for interaction.

The benefits of a two-tier architecture include ease of access to information, low upfront infrastructure costs, and high performance for a limited number of workstations.

However, the downside is this model isn't flexible enough to add more clients and software, and expensive middleware is needed to integrate the system with other components.

Any change or modification in the database also has an impact on applications (or clients). Another drawback is the implementation of business function modules depends a lot on the database servers or client application's implementation.



Three-Tier ERP Architecture



Presentation Layer



Application Layer



Database Layer

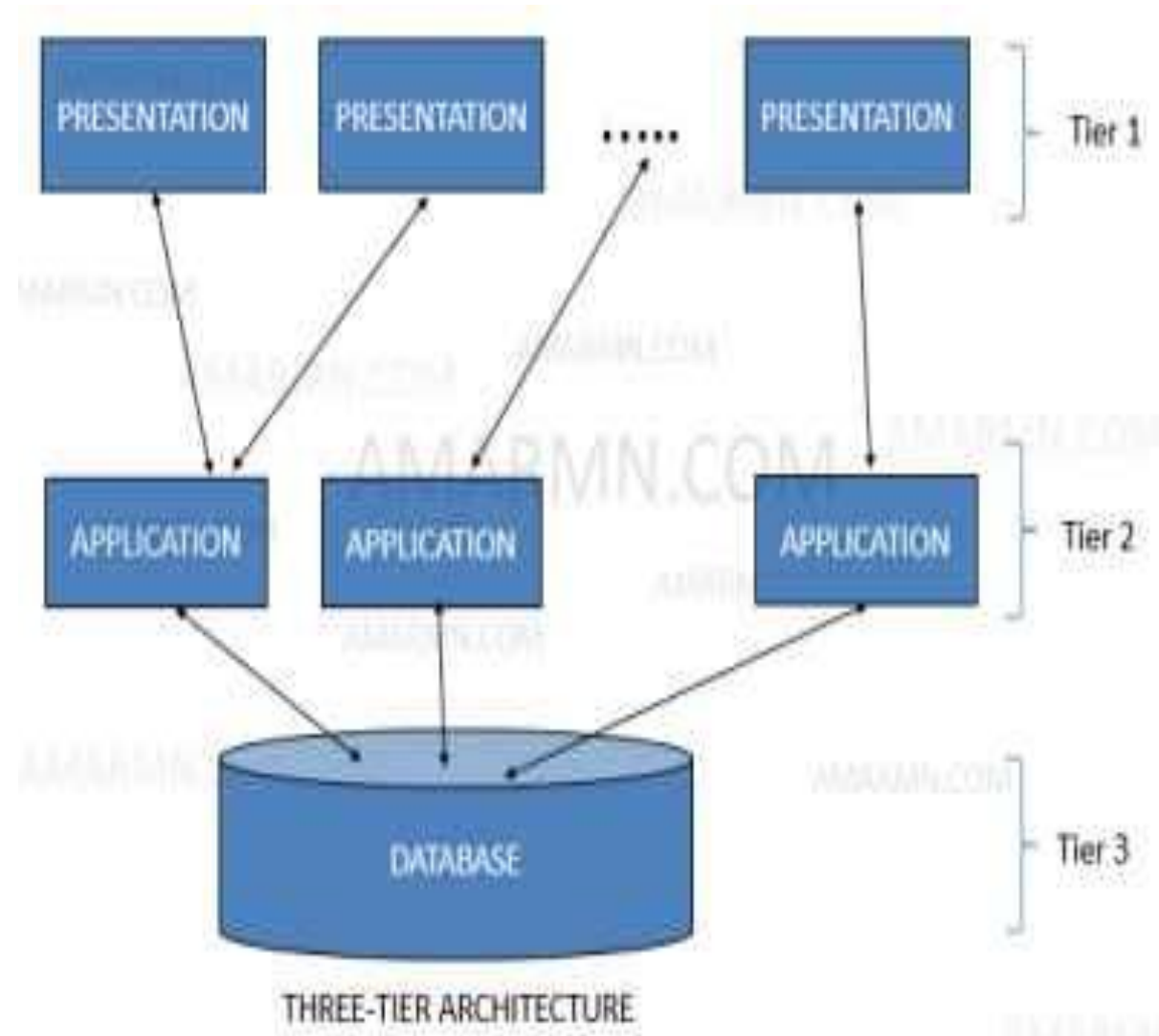
System Architecture Models of ERP – Three-Tier

A three-tier ERP architecture model consists of three layers: the presentation layer, the application layer, and the database layer. Each has a specific function within the ERP system.

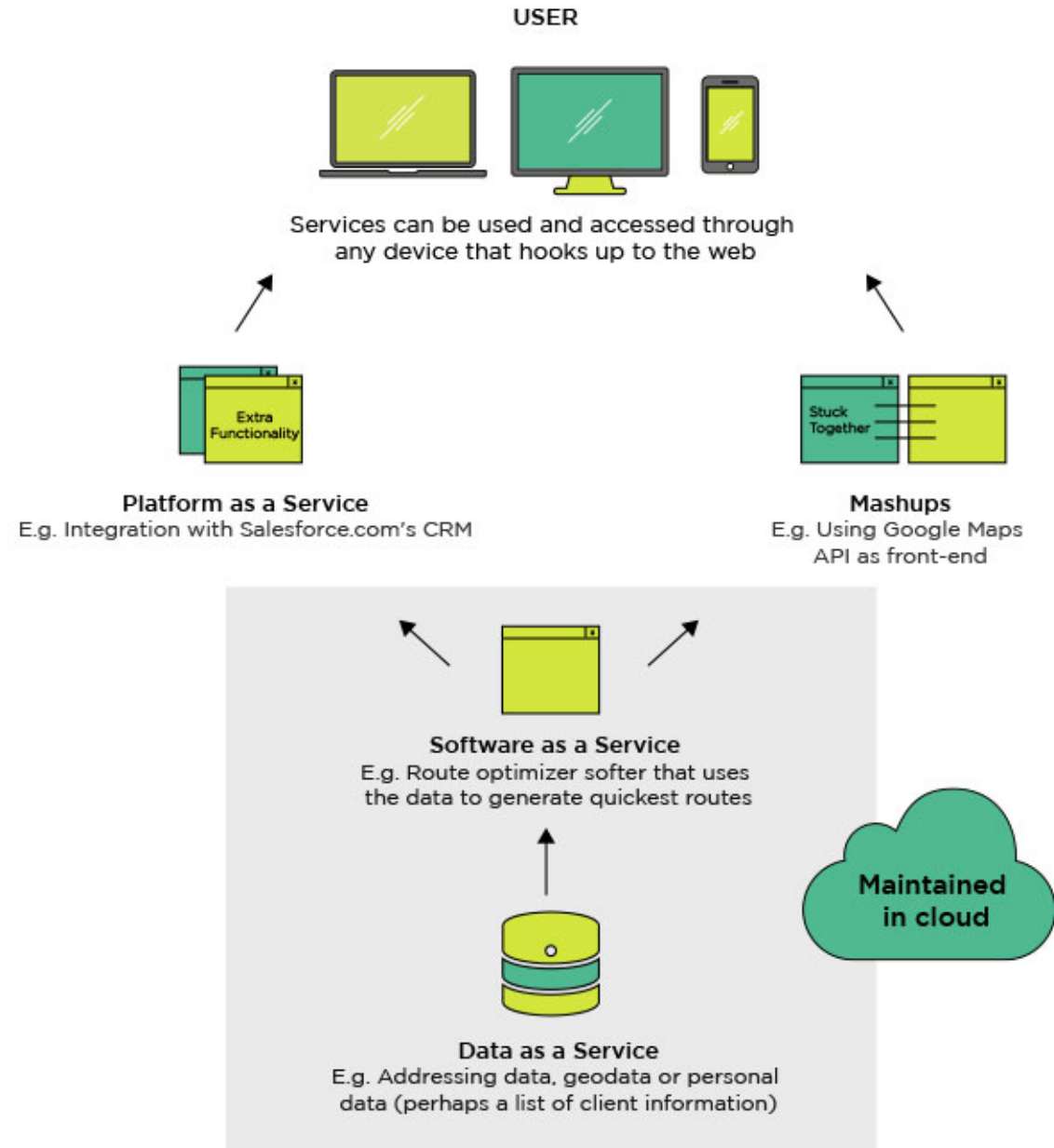
- **Presentation layer:** Provides the interface that allows application users to access and interact with an ERP system through a web browser or desktop app.
- **Application layer:** Handles the core functionalities and operations of an ERP system, such as executing business rules and processing data.
- **Database layer:** Serves as the repository where data from the ERP system is stored, organized, and accessed from a central database.

The separation and layout of layers means that each can scale independently. It also allows companies to customize and update individual layers without affecting the entire system. This configuration facilitates integration with external systems and applications.

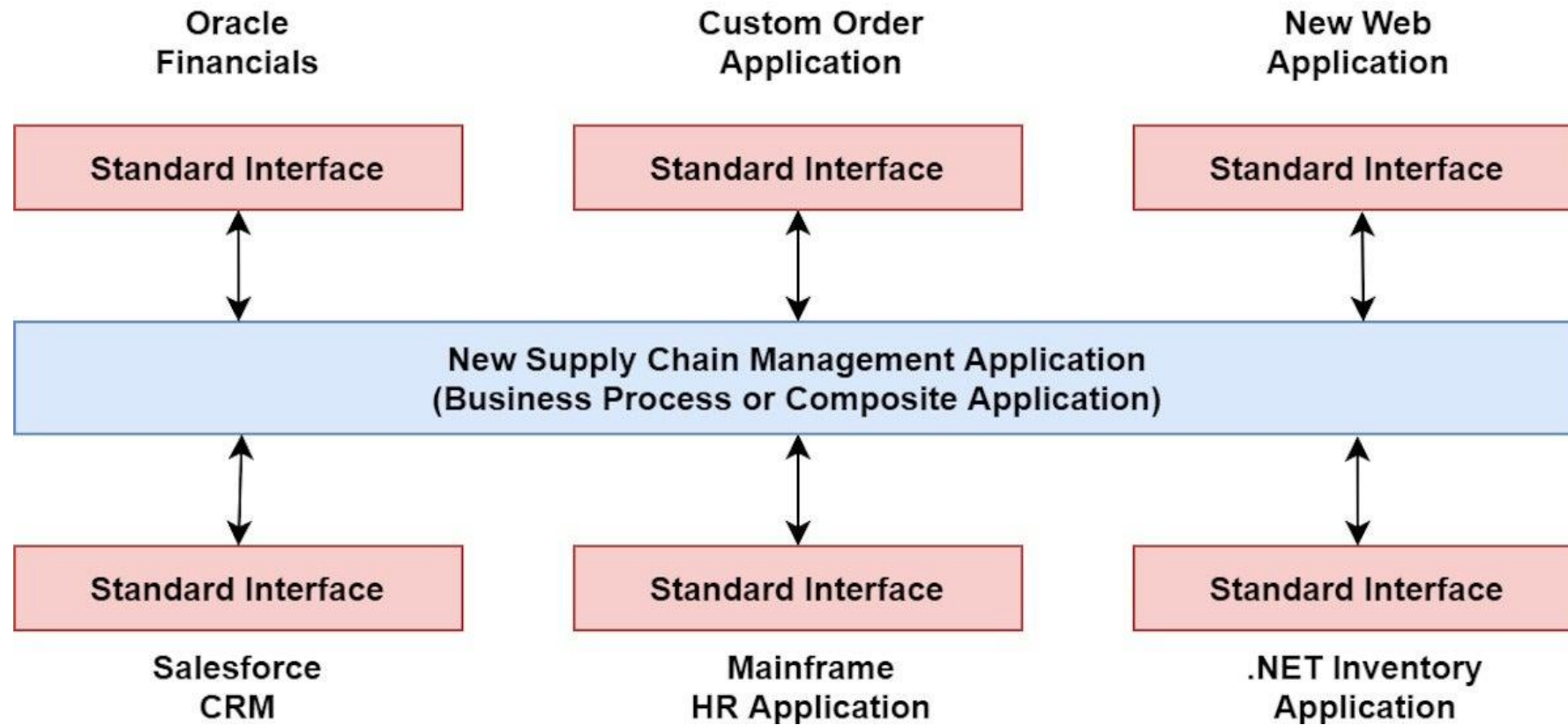
- The flexibility of this ERP architecture model allows it to meet the business requirements of all types of organizations, from multinational enterprises to manufacturing facilities and service-based companies.



Service-Oriented Architecture



System Architecture Models of ERP – SOA



Service Oriented Architecture (SOA) is a solution for making two applications communicate with each other by a collection of services. A service is self-independent, does not depend on the language, on which the consumer application is written. This is how the concept of the Service Oriented Architecture revolves, the consumer application will send the request to the service provider and in return, the response will be received by the consumer. The connections between the service provider and the service consumer will be done through the web services, which transfers the request and responding with that the data which is being requested.

Summary

- History
- Types of ERP
 - Build type
 - Deployment Model
 - Industry Specifications
 - Tiers
- System Architecture Models